

OfferPilot AI

Project Documentation and Software Design Document

A startup-ready interview intelligence platform for measurable candidate readiness, AI evaluation, live coding, resume analysis, learning roadmaps, and analytics.

Document Type	Software Company Design Document
Version	1.1 - OfferPilot AI Rename
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Project Overview

OfferPilot AI combines interview practice, answer evaluation, live coding, resume intelligence, analytics, and learning recommendations into one SaaS-ready product experience.

Architecture

The platform uses a modular React frontend, FastAPI backend, PostgreSQL persistence, Alembic migrations, Redis support, Docker Compose runtime, and provider abstractions for future AI integrations.

Backend Design

Backend code is organized by routers, schemas, services, repositories, models, core configuration, middleware, logging, validation, and exception handling. Environment variables use the OFFERPILOT_AI_ prefix.

Frontend Design

The frontend includes a premium OfferPilot AI landing page, responsive dashboard, dark mode, analytics workspace, live coding workspace, resume analyzer workspace, and branded favicon/logo placeholders.

Brand System

Primary color: Orbit Navy #0B1220. Secondary color: Signal Teal #14B8A6. Accent color: Offer Gold #F59E0B. The UI uses Inter, 8px card radius, restrained shadows, lucide icons, and light/dark theme tokens.

Database Design

PostgreSQL models support users, interviews, questions, answers, reports, learning roadmaps, leaderboard, sessions, interview history, evaluations, code submissions, and resume analyses.

Security

Security includes JWT authentication, refresh token rotation, password hashing, protected routes, role management, session management, trusted hosts, CORS allowlists, and production secret validation.

Deployment

The project runs through Docker Compose with API, frontend, PostgreSQL, and Redis services. Production config supports immutable images, migrations, health checks, and externalized secrets.

Commercial Vision

OfferPilot AI can serve job seekers, bootcamps, universities, recruiting agencies, coaching businesses, enterprise talent teams, and EdTech platforms through Free, Pro, Team, Enterprise, and API packaging.

Professional Conclusion

The renamed project now presents as a funded SaaS-style product with cohesive naming, branded assets, scalable architecture, production configuration, and a premium landing experience.

Area	Technology
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Frontend	React, TypeScript, Vite, Recharts, CodeMirror, lucide-react
Backend	FastAPI, Pydantic, SQLAlchemy, Alembic, Uvicorn
Data	PostgreSQL, JSONB, Redis
Runtime	Docker Compose, Nginx, health checks, environment files
Quality	pytest, API tests, integration tests, frontend TypeScript build